

The Growing Impact of Smart Cities

By Mohammad Shahidehpour and Mehdi Ganji

IT IS OUR PLEASURE TO PRESENT THIS SPECIAL ISSUE of *IEEE Electrification Magazine* on smart cities. This issue presents eight feature articles and two columns that discuss smart city technologies and applications.

The issue opens with the "Technology Leaders" column "Grid Modernization Is Paving the Way for Building Smarter Cities" by Aleksi Paaso, Daniel Kushner, Shay Bahramirad, and Amin Khodaei. The authors present an overview of the smart city concept and discuss potential opportunities for electric utilities and how they can become major players in implementing smart city programs. The article explains the ongoing initiatives pursued by ComEd in Chicago with a focus on smart grids as an enabler of smart city development.

In the first feature article, "Smart Cities for a Sustainable Urbanization," authors Mohammad Shahidehpour, Zhiyi Li, and Mehdi Ganji, point out that a smart city is an urban center that integrates a variety of innovative solutions to improve infrastructural performances and achieve a sustainable urban development. The urban infrastructures in many cities suffer from a series of critical issues, including capacity insufficiency, functional deterioration and deferred maintenance, and technological obsolescence that place legacy infrastructures unprepared for providing better civil services and promoting a higher quality of life and socioeconomic competitiveness. In the future, the authors explain, a significant increase in urban population will place massive pressure on constrained city infrastructures (e.g., public and private transportation and mobility), pose various types of concerns for peoples' lives, and, potentially, lead to significant economic losses and lack of preparedness in critical and disastrous circumstances. The adoption of smart city solutions represents a key factor in the consumption of resources for improving the efficiency of services and meeting individual citizens' needs as urban populations grow and resources become scarcer.

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The article “How to Create a Smart City,” by Don Reeves states that networking platforms that deliver reliable connectivity (always a top priority) must be established for smart cities to be successful. Though choosing the right network platform can be a challenge, the author describes how cities can deploy multiple, singular-use platforms, just as they already do for voice and data. But building and managing these networks adds cost, creates complexity, and inhibits interoperability. To drive costs down, it is necessary to ensure the ability to do distributed computing on the network. Otherwise, the network simply collects data and delivers it to the cloud, which means that, over time, the network will become more congested and cloud compute and storage costs will grow unbounded. However, if compute can be done wherever sensors are deployed and sensors can talk with other local sensors, then both network congestion and cloud cost issues will be avoided.

The next article, “Integrated Energy Management System” by Hongbin Sun, Qinglai Guo, Boming Zhang, Wenchuan Wu, Bin Wang, Xinwei Shen, and Jianhui Wang, discusses how to overcome the difficulties in reforming the energy industry and realize a truly interconnected energy system. An energy Internet (EI) is the most important topic in innovative energy research circles, and, in coming decades, it could break down the barriers between different energy sections in traditional energy ecosystems so that the planning and operation of all types of energy flows, e.g., electricity, heating, cooling, natural gas, transportation, and other massive infrastructures can be optimized together. EI will change the methods of energy production and transmission as well as consumption patterns. EI is expected to become an important approach to promoting energy transformation, improving energy efficiency and conservation, reducing emissions, and developing sustainable energy, which will have a profound impact on other industries and society as a whole.

The article, “Smart City Solutions and Applications in China,” by Jinda Zhu, Houtao Sun, Di Wu, Luyao Chen, Jian Luo, Zhuofei Yu, Yonghua Chen, Jianfu Ni, and Kun Huang, focuses on the evolution of smart cities. The authors point out that smart cities are more than a simple combination of information and intelligence; rather, they are industrial ecosystems designed to enhance people’s livelihood and respond to urban development, industrial upgrades, and

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changes to the global economy. The authors assert that smart cities must integrate the planning and operation of resources to facilitate interaction among city infrastructures, thus providing a roadmap for economic development while acknowledging the interdependency of industrial chains. The smart city idea was initiated in China around 2010, with more than 26 major cities promoting smart cities since then. At present, hundreds of cities in China have witnessed a climax of smart city development. In 2014, the Chinese government released the National New Urbanization Plan (2014–2020), which mentioned smart city development as one of the three important directions for a new city. In June 2014, the National Development and Reform Commission released a white paper, “China Smart Grid and Smart City Development Report,” and the Chinese premier further detailed a concept for building smart cities and improving the living conditions of major Chinese cities.

In “IoT-Enabled Humans in the Loop for Energy Management Systems,” Mona Bisadi, Alireza Akrami, Saeed Teimourzadeh, Farrokh Amini-far, Mehdi Kargahi, and Mohammad Shahidehpour state that conventional building energy management system (BEMS) models can save energy satisfactorily but are not controlled to adapt to building occupants’ behavior to increase productivity and comfort. The authors propose a new BEMS concept that can promote the role of customer participation in optimizing energy consumption. Here, Internet of Things (IoT) technology is integrated with conventional BEMS to cope with big data collection and management challenges in residential and commercial buildings and prepare required framework transitions toward smart buildings. The proposed BEMS framework can help lower energy consumption, reduce environmental pollution, and maximize building

occupants’ performances. The authors detail how a BEMS model embeds a combination of the IoT and HiTLCPS, in which humans play a sensory role and their situational awareness promotes the adaptability of a BEMS controller. The required framework for the massive data transmission is prepared by the IoT, considering occupants’ physical and physiological states that are handled by HiTLCPS.

In, “Coupled Cyber and Physical Systems,” authors Xiaonan Lu, Bo Chen, Chen Chen, and Jianhui Wang explain how smart city revolutions are driven by the rapid pace of urbanization, the fast deployment of IoT technologies, an unprecedented volume of data, and associated data analytic technologies. In smart cities, data are gathered from numerous sensors, computer processes, and applications that are distributed throughout various infrastructures. Multiple city infrastructures (e.g., power facilities, water, gas, oil, transportation, and communication activities) generate data streams by sensors and multistream data flows that are routed through dedicated networks and processed by various applications designed for smart cities, which are subjected to cyberphysical outages. In this paradigm, realizing that every city’s infrastructure is designed to serve people living in the physical world, simply cutting off the affected data sources in cyberphysical incidents could be no better than doing nothing. Therefore, mitigation strategies must take into consideration the interdependency of cyber and physical layers, identify critical city elements that ought to survive major events, and aim to minimize the impact of cyber incidents in extreme circumstances.

Our next article, “Distribution Management System for the Grid of the Future,” is by Farrokh A. Rahimi and Sasan Mokhtari. The authors assert that electric utilities in smart cities are facing both operational and

business model challenges due to a number of factors, primarily concerned with increased distributed energy resource (DER) penetration. Operational issues, the authors say, are mostly related to a lack of visibility and controllability of customer/prosumer owned and operated DERs, as well as their variability and unpredictability. The business model issues are attributed to reduced sales due to customer-side generation. Regulatory provisions are needed so that utilities can provide and charge for new products and services to fill in the revenue gaps, and prosumers can tap into new sources of revenue to cover their investments and make new investments in consumer/prosumer side assets. With proper regulatory provisions, trends such as grid divorce can find a new venue that the authors refer to as *graceful grid divorce*, whereby the prosumers can opt out of some distribution grid services they do not need and pay for those they do, thereby requiring the unbundling of distribution services. For example, the distribution utility can provide and charge for transport service as a combination of connection cost and pay-as-you-go transport cost. The article discusses the functional requirements of a transactive distribution management system, which provides illustrative examples and use cases along with preliminary results on this critical topic for managing smart cities.

The final feature article, “A Coordinated Utility Emergency Response,” comes to us from James B. Marean and William G. Gale. The authors state that the ability to provide natural gas utility companies and organizations that rely on natural gas service with the ability to assess threats and vulnerabilities, understand interdependencies and their cascading effects, and hasten recovery is a critical feature of the situational awareness system developed by the team led by the Gas Technology Institute. However, major disaster

events require response by more than one emergency responder, utility, or agency. In this regard, the utility situational awareness system (USAS) system provides a data exchange methodology for critical natural gas infrastructure operators and government agencies ensuring a coordinated response, allowing them to share information, provide situational awareness, and assist with decision support during a disaster. Based on the status of each structure while maintaining confidentiality of the distribution systems for individual electric, gas, water, phone, and cable utilities, the USAS will provide almost immediate status information to all responders, allowing them to know the extent of the event, the availability of infrastructure, such as accessibility to roads, bridges (open or closed), electric status (on or off) and the like—for each structure within the effected area. A coordinated response can have cascading impacts throughout the energy supply chain, providing a comprehensive picture of the state of all utilities, and optimizing restoration and the use of critical resources.

Finally, in the “Viewpoint” column, “Four Steps to Smart City Success,” Jesse Berst points out that most cities fail badly at thinking big. This failure puts their entire smart city effort at risk. The lack of big-picture thinking, i.e., holistic thinking and systems thinking, has devastating consequences, including choosing the wrong priorities, creating piecemeal solutions that are trapped in departmental silos, reinventing wheels (and paying for them multiple times), and failing to capture synergies—opportunities to share infrastructure, share costs, and share data. However, holistic thinking is only half the story; the other half is a customer-centric mindset. Many governments still think of citizens as inhabitants, not as customers. For one thing, serving citizens is too limiting, as cities also serve tourists, day workers, businesses,

investors, city staff, and many others. For another, it is a useful reminder that cities are competing for those customers with all the other cities that want to attract jobs, talent, and

tourists. The author discusses four comprehensive steps that provide a competitive advantage for establishing smart cities.



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